Ron Kitainik

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Education

B.A.Sc Engineering Physics, Minor in Honours Mathematics — Sep 2022 — 2027
University of British Columbia • 3.7/4.0 GPA • Co-op Student • Available from May-Aug 2026 — *Vancouver, BC*

Teams & Projects

Gas Thruster Lunar Drone — Sep 2025 – Present
Engineering Physics Capstone

- Leading physical modeling and electrical design for a cold gas thruster drone for potential lunar deployment
- Developing Simulink model with rotational kinematics and propellant tracking for early hardware-in-the-loop tests
- Prototyping embedded ESP32-based valve driver circuit with sensor integration (load cell, pressure transducer and IMU)

Hardware Integration Lead — Sep 2024 – Dec 2025
UBC Orbit Design Team

- Designed a custom flexible harness in Altium to transmit RF signals and sensor data onboard ALEASAT, an imaging satellite
- Configured a polyamide/epoxy layer stack to maximize PCB lifetime in orbital vacuum and thermal conditions
- Documented testing procedures in OpenProject to ensure compliance with European Space Agency standards

Arm Mounted Rocket Launcher — Nov 2025

- Built a device out of cardboard for the BoxBots hackathon that safely aimed and fired class A model rockets
- Directed circuit assembly and MCU logic; integrated camera, servo, and ignition wire into the control system

Simulated Self-Driving Car — Jan – Apr 2025

- Implemented an RL algorithm to control hazard avoidance and autonomous navigation for a simulated car
- Developed an MLP neural network to interpret text from virtual camera data, finishing with 98% accuracy

Work Experience

Mission Management Intern — Sep – Dec 2025
Canadian Space Agency — *Longueuil, QC*

- Optimized physics for a Python simulator to generate atmospheric data for the TICFIRE climate research satellite; improved computation speed by 2.6×, and added compatibility for modern radiative transfer models
- Validated simulator performance in laboratory using radiometer and microbolometer, reaching 92% accuracy
- Led contractor communication through kick-off and phase A; built automated project dashboards with Power BI

Quant Developer — May – Aug 2025

- Constructed and backtested several trading strategies using QuantConnect to manage a small company's finances, including a mean reversion model, market news sentiment analyzer, and neural network trained on OHLC data
- Deployed a paper-traded algorithm with a Sharpe ratio of 1.4 for validation prior to live market deployment

Survey Operator — May – Aug 2023
City of Vancouver — *Vancouver, BC*

- Conducted layout and construction surveys for major water main replacements using total station and laser scanner
- Drafted and revised engineering drawings, utility plans, and legal documents in AutoCAD and Civil 3D

Skills

- **Electrical/Systems:** Altium, KiCad, SPICE, Simulink, control systems, embedded systems, RF design, bench tools
- **Software & Development:** Linux, Bash, Python, TensorFlow, OpenCV, MATLAB, Java, C/C++, microcontrollers
- **Design & Fabrication:** SolidWorks, AutoCAD, Fusion 360, PLCs, 3D printing, waterjet cutting, CNC, shop tools